

Intelligent Course Recommendation System

Power BI Dashboard — Technical Documentation

Coursera · Udemy · Udacity

1. Project Overview

The Intelligent Course Recommendation System is a data engineering and analytics project that collects, transforms, and visualizes online course data from three major e-learning platforms: Coursera, Udemy, and Udacity. The goal is to help learners identify the best courses based on multiple quality metrics.

2. Data Sources

Platform	Content Type	Key Data Points
Coursera	Courses & Specializations	Rating, Enrolled Students, Duration, Price
Udemy	Single Courses	Rating, Reviews, Price, Duration
Udacity	Nanodegrees & Programs	Rating, Enrollment, Duration, Price

3. Data Pipeline & Connection Setup

3.1 DuckDB ODBC Driver Installation

To connect Power BI to the DuckDB database, the DuckDB ODBC Driver was installed on the local machine:

1. Downloaded the DuckDB ODBC Driver from: <https://duckdb.org/docs/api/odbc>
2. Ran the installer and completed the setup on the Windows machine
3. Verified the driver in Windows ODBC Data Source Administrator (odbcad32.exe)

3.2 Configuring the Data Source (DSN)

4. Opened Windows ODBC Data Source Administrator (Start → search 'ODBC Data Sources')
5. Clicked 'Add' under the 'User DSN' tab

6. Selected 'DuckDB Driver' from the list
7. Set the database path to the local dbt_local.duckdb file
8. Saved the DSN configuration

3.3 Connecting Power BI to DuckDB

9. Opened Power BI Desktop
10. Clicked Home → Get Data → More
11. Searched for 'ODBC' and selected it
12. Selected the DuckDB DSN from the dropdown
13. Clicked Connect
14. In the Navigator, selected all required tables: fact_offerings, dim_platform, dim_domain, dim_level, dim_language, dim_offering_type, dim_instructor, dim_date, dim_offering
15. Clicked Load to import all tables into Power BI

3.4 Data Model Relationships

From Table	To Table	Cardinality
fact_offerings	dim_platform	Many to One (*:1)
fact_offerings	dim_domain	Many to One (*:1)
fact_offerings	dim_level	Many to One (*:1)
fact_offerings	dim_language	Many to One (*:1)
fact_offerings	dim_offering_type	Many to One (*:1)
fact_offerings	dim_instructor	Many to One (*:1)
fact_offerings	dim_date	Many to One (*:1)
fact_offerings	dim_offering	Many to One (*:1)

4. Data Model (Star Schema)

The data model follows a Star Schema design with one central fact table surrounded by dimension tables.

4.1 Fact Table — fact_offerings

Column	Data Type	Description
offering_sk	INT	Surrogate key linking to dim_offering
platform_id	INT	Foreign key to dim_platform

domain_id	INT	Foreign key to dim_domain
level_id	INT	Foreign key to dim_level
language_id	INT	Foreign key to dim_language
offering_type_id	INT	Foreign key to dim_offering_type
instructor_id	INT	Foreign key to dim_instructor
date_id	INT	Foreign key to dim_date
price	DECIMAL	Course price
rating	FLOAT	Average rating (0-5)
review_count	INT	Number of reviews
enrolled_students	INT	Total enrolled students
duration_hours	FLOAT	Course duration in hours
popularity_score	DECIMAL	rating x LN(enrolled_students + 1)
engagement_rate_pct	DECIMAL	(review_count / enrolled_students) x 100

4.2 Dimension Tables

Table	Key Columns	Purpose
dim_offering	offering_sk, title, course_url, skills	Course details and attributes
dim_platform	platform_id, platform_name	Coursera / Udemy / Udacity
dim_domain	domain_id, domain_name	Course subject area / track
dim_level	level_id, level_name	Beginner / Intermediate / Advanced
dim_language	language_id, language_name	Course language
dim_offering_type	offering_type_id, offering_type	Course / Certificate / Nanodegree
dim_instructor	instructor_id, instructor_name	Course instructor
dim_date	date_id, full_date, month, year	Date dimension for time analysis

5. Calculated Metrics

Metric	Formula	Purpose
Popularity Score	rating x LN(enrolled_students + 1)	Balances quality and reach
Weighted Rating	rating x LN(review_count + 1)	Rating adjusted by review volume

Engagement Rate %	$(\text{review_count} / \text{enrolled_students}) \times 100$	Measures learner engagement
Cost Per Hour	$\text{price} / \text{duration_hours}$	Pricing efficiency
Price Category	Free / Cheap / Medium / Expensive / Premium	Price segmentation
Duration Category	Short / Medium / Long / Extensive	Duration segmentation

6. Dashboard Structure

6.1 Cover Page

- Project title: Intelligent Course Recommendation System
- Platform names: Coursera · Udemy · Udacity

6.2 Dashboard 1 — Course Recommendation

Filters (Slicers):

- Domain Name · Level Name · Platform Name · Price Category · Duration Category

KPI Cards:

- Avg Rating · Total Enrolled · Cost Per Hour · Weighted Rating

Visuals:

- Bar Chart — Top 3 courses by Weighted Rating (Top N filter = 3)
- Table — Top 3 courses: Title, Platform, Rating, Duration, Weighted Rating, Price

6.3 Dashboard 2 — Course Analytics

Filters (Slicers):

- Domain Name · Level Name · Platform Name

Visuals:

- Pie Chart — Platform Distribution by enrolled students
- Bar Chart — Popularity Score by Domain Name
- Column Chart — Enrollment by Offering Type

7. Standardized Domain Values

Domain	Domain
Data Analytics	Artificial Intelligence
Data Science	Machine Learning

Programming	Web Development
Cyber Security	UI/UX Design
DevOps	Cloud Computing
Networking	Embedded Systems
Software Engineering	Business & Management
Other	

8. Standardized Filter Values

Filter	Values
Platform	Coursera · Udemy · Udacity
Level	Beginner · Intermediate · Advanced · All Levels
Offering Type	Single Course · Certificate · Specialization · Nanodegree · Diploma
Duration	Short (<5h) · Medium (<20h) · Long (<50h) · Extensive (50h+)
Price	Free · Cheap (<\$500) · Medium (<\$2000) · Expensive (<\$5000) · Premium
Price Model	One Time · Subscription

9. Tools & Technologies

Tool	Purpose
DuckDB	Local data warehouse / database
DuckDB ODBC Driver	Enables connection between DuckDB and Power BI
dbt (Data Build Tool)	Data transformation and modeling
Power BI Desktop	Dashboard creation and visualization
Python	Data loading and pipeline scripting
GitHub	Version control and collaboration